RPractical_Fantilanan

Glen Mark F. Fantilanan

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R Practical

```
#A1

v <- seq(5, 50, by = 5)
length(v)   # 10
```

```
## [1] 10
```

```
sum(v)      # 275
```

```
## [1] 275
```

```
mean(v)     # 27.5
```

```
## [1] 27.5
```

```
#A2

w <- rep(c(1, 2, 3), times = 4)
z <- rep(c(1, 2, 3), each = 4)
print(w)    # 1 2 3 1 2 3 1 2 3 1 2 3
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
print(z)    # 1 1 1 1 2 2 2 2 3 3 3 3
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

```
#A3

17 %% 5        # 2   (remainder)
```

```
## [1] 2
```

```r
17 %/% 5       # 3   (integer division)
```

```
## [1] 3
```

```r
2^10           # 1024
```

```
## [1] 1024
```

```r
sqrt(144)      # 12
```

```
## [1] 12
```

```r
round(pi, 3)   # 3.142
```

```
## [1] 3.142
```

```r
#A4

a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)

a + b          # 3 7 11 15
```

```
## [1]  3  7 11 15
```

```r
a * b          # 2 12 30 56
```

```
## [1]  2 12 30 56
```

```r
a > b          # TRUE TRUE TRUE TRUE
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a * b)     # 100
```

```
## [1] 100
```

```r
#A5

paste("Student", 1:3, sep = "-")
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```r
# "Student-1" "Student-2" "Student-3"

class(42)      # "numeric"
```

```
## [1] "numeric"
```

```r
class(42L)     # "integer"
```

```
## [1] "integer"
```

```r
class("42")    # "character"
```

```
## [1] "character"
```

```r
class(41i)     # "complex"
```

```
## [1] "complex"
```

```r
class(TRUE)    # "logical"
```

```
## [1] "logical"
```

```r
#B1

students <- data.frame(
  id         = 1:8,
  name       = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
  sex        = c("F", "M", "M", "F", "F", "M", "F", "M"),
  section    = c("A", "B", "A", "C", "B", "C", "A", "B"),
  midterm    = c(78, 85, 92, 67, 74, 88, 95, 59),
  final      = c(82, 79, 96, 71, 80, 91, 93, 65),
  attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
  stringsAsFactors = FALSE
)
```

```r
#B2

str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```r
dim(students)     # 8 7
```

```
## [1] 8 7
```

```r
names(students)   # "id" "name" "sex" "section" "midterm" "final" "attendance"
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```r
summary(students)
```

```
##        id              name               sex               section     midterm
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                           Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

```r
#B3

students$average <- round((students$midterm + students$final) / 2, 1)
students$average
```

```
## [1] 80.0 82.0 94.0 69.0 77.0 89.5 94.0 62.0
```

```r
# 80.0 82.0 94.0 69.0 77.0 89.5 94.0 62.0
```

```r
#B4

students$passed <- students$average >= 70 & students$attendance >= 75
students$passed
```

```
## [1]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE
```

```r
# TRUE TRUE TRUE FALSE TRUE TRUE TRUE FALSE
```

```r
#B5

students$standing <- ifelse(students$average >= 90, "Honors",
                     ifelse(students$average >= 75, "Good", "Needs Support"))
students[, c("name", "average", "standing")]
```

```
##    name average      standing
## 1 Alice    80.0          Good
## 2 Bruno    82.0          Good
## 3  Chen    94.0        Honors
## 4  Dana    69.0 Needs Support
## 5  Elif    77.0          Good
## 6 Farid    89.5          Good
## 7 Grace    94.0        Honors
## 8  Hiro    62.0 Needs Support
```

```r
#B6

classify <- function(score) {
  if (score >= 85) {
    "Distinction"
  } else if (score >= 70) {
    "Merit"
  } else if (score >= 50) {
    "Pass"
  } else {
    "Fail"
  }
}

result <- sapply(c(90, 73, 55, 40), classify)
print(result)
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

```r
# [1] "Distinction" "Merit"       "Pass"        "Fail"
```

```r
#C1

new_students <- data.frame(
  id         = c(9, 10),
  name       = c("Ines", "Jamal"),
  sex        = c("F", "M"),
  section    = c("C", "A"),
  midterm    = c(81, 70),
  final      = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)

new_students$average  <- round((new_students$midterm + new_students$final) / 2, 1)
new_students$passed   <- new_students$average >= 70 & new_students$attendance >= 75
new_students$standing <- ifelse(new_students$average >= 90, "Honors",
                         ifelse(new_students$average >= 75, "Good", "Needs Support"))

students_all <- rbind(students, new_students)
students_all[9:10, c("name", "average", "passed", "standing")]
```

```
##     name average passed      standing
## 9   Ines    82.5   TRUE          Good
## 10 Jamal    69.0  FALSE Needs Support
```

```r
#C2

quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz = quiz)
class(students_all)   # "data.frame"
```

```
## [1] "data.frame"
```

```
#C3

m <- cbind(midterm = students_all$midterm, final = students_all$final)
m <- rbind(m, mean = colMeans(m))
m
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

```
#C4

# rbind() stacks data frames on top of each other, but the column names have to
# match. If they don't, R gives an error: "names do not match previous names".
# The assignment says 2-3 sentences, so I used tryCatch() to catch the error
# so the code doesn't crash and just prints a message instead.

bad <- new_students
names(bad)[2] <- "student_name"   # changed "name" to "student_name" so it doesn't match students

tryCatch(rbind(students, bad),
         error = function(e) {
           message("rbind failed: ", conditionMessage(e))
           NULL
         })
```

```
## rbind failed: names do not match previous names
```

```
## NULL
```

```
# Output:
# rbind failed: names do not match previous names
# NULL
```

```
#D1

students_all$name
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
students_all[["name"]]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
students_all[, "name"]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
# all three: "Alice" "Bruno" "Chen" "Dana" "Elif" "Farid" "Grace" "Hiro" "Ines" "Jamal"
```

```r
#D2

students_all[2:5, c("name", "midterm", "final")]
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

```r
#D3

students_all[students_all$attendance > 90, ]
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good    8
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good    8
```

```r
#D4

subset(students_all,
       sex == "F" & section %in% c("A", "C") & final >= 75,
       select = c(name, section, final))
```

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

```r
#D5

students_all$name[which.max(students_all$average)]   # "Chen"
```

```
## [1] "Chen"
```

```r
students_all$name[which.min(students_all$average)]   # "Hiro"
```

```
## [1] "Hiro"
```

```r
#D6

sorted <- students_all[order(-students_all$average), ]
head(sorted, 3)
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 6  6 Farid   M       C      88    91         85    89.5   TRUE     Good    9
```

```r
#E1

students_all$section <- factor(students_all$section)
students_all$sex     <- factor(students_all$sex)

levels(students_all$sex)       # "F" "M"
```

```
## [1] "F" "M"
```

```r
levels(students_all$section)   # "A" "B" "C"
```

```
## [1] "A" "B" "C"
```

```r
#E2

table(students_all$sex)
```

```
##
## F M
## 5 5
```

```r
# F M
# 5 5

table(students_all$section)
```

```
##
## A B C
## 4 3 3
```

```r
# A B C
# 4 3 3

table(students_all$sex, students_all$section)
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

```r
#E3

as.integer(students_all$sex)
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

```r
# 1 2 2 1 1 2 1 2 1 2

# The integers are the codes R uses to store the factor. R doesn't keep "F" and "M"
# as text, it saves them as numbers and keeps a list of levels to match them to.
# Level 1 is "F" and level 2 is "M" (alphabetical order), so 1 = F and 2 = M.
# For example, Alice is F so she gets 1 and Bruno is M so he gets 2.
```

```r
#E4

students_all$sex_label <- factor(students_all$sex,
                                 levels = c("F", "M"),
                                 labels = c("Female", "Male"))
levels(students_all$sex_label)   # "Female" "Male"
```

```
## [1] "Female" "Male"
```

```r
#E5

summary(students_all$sex_label)
```

```
## Female   Male
##      5      5
```

```r
# Female   Male
#      5      5

mean(students_all$average[students_all$sex_label == "Female"])   # 80.5
```

```
## [1] 80.5
```

```r
mean(students_all$average[students_all$sex_label == "Male"])     # 79.3
```

```
## [1] 79.3
```